

Zachary Selzman

+1 (801) 419-9640 | Salt Lake City, UT | zachselzman@gmail.com | [linkedin.com/in/zach-s-87ba23290](https://www.linkedin.com/in/zach-s-87ba23290) | zachsstms.github.io

WORK EXPERIENCE

SULI Intern

June 2025 — August 2025

National Laboratory of the Rockies, US Department of Energy

Boulder Colorado

- Worked as a undergraduate engineering intern at the National Wind Technology Center for the Field Test and Validation Group
- Performed data analysis on small distributed wind turbines to determine the effects of obstacle wakes on power performance in accordance with IEC standards
- Completed various deliverables, including a full-scale research paper and presentation

Aerodynamics Team Member

September 2024 — Present

Cal Poly Wind Power

San Luis Obispo, CA

- Work with a small team developing the blades, nosecone, and nacelle cover for the club wind turbine, optimizing for successful operation at the DOE Collegiate Wind Power competition (CWC)
- Use Qblade and Xfoil-based panel codes to quantify airfoil power output for a range of wind speeds and tip-speed ratios
- Perform testing and validation of the turbine in a wind tunnel, and blade structural testing

Instructional Student Leader

September 2024 — June 2025

California Polytechnic State University

San Luis Obispo, CA

- Ran student tutor sessions in Physics, Statics, and Multivariable Calculus
- Assisted with test preparation, homework help, and general lectures

EDUCATION

California Polytechnic State University

San Luis Obispo, CA

Bachelor's of Science, Aerospace Engineering

Sep 2023 — Expecting June 2027

- Dean's List Recipient, member of the AIAA, Wind Power, and Alpine Clubs

PROJECTS

Air Motor

California Polytechnic State University

San Luis Obispo, CA

- Designed and machined parts of an air motor on a lathe and CNC mill in addition to outsourced casts

Screwdriver

California Polytechnic State University

San Luis Obispo, CA

- Designed and machined a screw driver with a modular tip for both flathead and phillips head bits

Aerodynamic Code Adventure

California Polytechnic State University

San Luis Obispo, CA

- Used `Waterlily.jl`, a Julia-based viscous flow solver to characterize the Strouhal number of unsteady flow around a cylinder for various Reynolds numbers
- Deployed various boundary conditions and flow visualization techniques

TECHNICAL SKILLS

Matlab, Python, Julia, Qblade, simFlow, Solidworks, AutoCAD, Ansys Fluent, Granta EDUpack Arduino, GD&T, Test Engineering

PERSONAL INTERESTS

Climbing, Running, Paragliding, Stall Aerodynamics, Aeroelasticity